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The stress has always been on flexing more and more computer muscle. However, a welcome shift in thinking is taking place. Researchers are now working on designing and building computing systems that are capable of running, healing and protecting themselves

Ganesh is fuming. His hard disk has crashed. He's struggling to retrieve his precious data. You can almost hear him swear: "Aaargh! One more time and I'll throw away this blasted box." This refrain is nothing new to a computer user. Even Richard Feynman, in his Lectures on Computation, summed up Von Neumann computers by saying, "the inside of a computer is as dumb as hell, but it goes like mad!" Never mind the context, but even as processors keep adding more hertz by the day, somebody's OS keeps playing truant, another's modem continues gargling to no end without a decent hand-shake. A third is forced to stare mindlessly at an 'invalid error'. Now, while geeks and vendors can give you sermons on how to solve these problems, that's not how lesser mortals would like a computer to behave. And that's no constructive way to spend our waking time either.

If only computers could behave like our bodies do by way of the Autonomic Nervous System (ANS) which, among other things, automatically handles basic functions such as breathing and digestion. It maintains the temperature at 98.6 degrees Fahrenheit and rids the body of heat by shuttling blood from the internal organs to the skin. The ANS tells the heart to maintain the right number of beats. If the blood pressure drops, the ANS changes the resistance of the veins because 70 per cent of circulating blood is in the veins. If only computers could act likewise and main-



ILLUSTRATIONS: Mahesh Benkar